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>92%	8×	<30°C	0
SYSTEM EFFICIENCY	LESS HEAT LOSS	OPERATING TEMP	RARE EARTH METALS

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CASE STUDY

HYDROGEN FOR HOSPITALS

INTEGRATION CASE STUDY

Version 2.0 · February 2026

EXECUTIVE SUMMARY

Large hospitals face increasing pressure to decarbonize operations, enhance energy resilience, and manage costs effectively, all while maintaining the highest standards of patient care. This document presents an investor-focused use case for integrating on-site green hydrogen production and utilization within a large hospital setting. Green hydrogen, produced via renewable-powered electrolysis, offers a unique, multifaceted solution by simultaneously generating medical-grade oxygen, providing emissions-free backup power, enabling energy cost optimization through peak shaving, supporting advanced medical research, and potentially offering novel therapeutic applications.

This integrated approach not only addresses environmental mandates (the hospital sector contributes **~4.4% of global CO_2 emissions** [1]) but also offers significant operational cost savings, enhanced supply chain security for critical resources like oxygen, and improved overall resilience against grid instability and commodity price volatility. For investors, this represents an opportunity to fund a transformative infrastructure upgrade that yields tangible financial returns, mitigates operational risks, and aligns with compelling ESG (Environmental, Social, and Governance) objectives within the vital healthcare sector.

1. INTRODUCTION: THE IMPERATIVE FOR SUSTAINABLE AND RESILIENT HEALTHCARE

The global healthcare sector, responsible for approximately **4.4% of net global carbon emissions** [1], is at a critical juncture. Hospitals are increasingly mandated to reduce their environmental footprint, with over **14,000 facilities worldwide pledging to halve emissions by 2030** [1]. Simultaneously, the absolute necessity for uninterrupted operations and resource availability — power, medical gases — demands robust and resilient infrastructure.

Green hydrogen, produced through water electrolysis powered by renewable energy, presents a compelling pathway to address these dual challenges. It allows hospitals to decarbonize significantly while enhancing operational self-

sufficiency and resilience [1]. This document outlines the strategic integration of green hydrogen systems in large hospitals, detailing key applications, financial considerations, and the long-term value proposition for stakeholders and investors.

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2. ON-SITE MEDICAL-GRADE OXYGEN PRODUCTION VIA ELECTROLYSIS

A primary output of water electrolysis, alongside hydrogen, is high-purity oxygen [2]. By installing a hospital-scale electrolyzer, a facility can generate its own medical-grade oxygen, significantly reducing or eliminating dependence on external bulk liquid oxygen suppliers.

This offers a critical advantage in supply security, particularly highlighted during crises like the COVID-19 pandemic, where hospital oxygen demand surged by approximately **50%**, straining conventional supply chains [2]. The Viamed San José hospital in Spain, for instance, utilizes a solar-powered electrolyzer to produce hydrogen for heating and captures the co-produced oxygen for medical use, enhancing self-sufficiency and reducing CO_2 emissions by **20 tons annually** [2].

To meet stringent medical standards (e.g., European Pharmacopeia mandates **≥93% purity** for on-site generated O_2 [3]), the oxygen undergoes filtration and continuous monitoring. Quality control and certification, adhering to medical gas regulations and Good Manufacturing Practices (GMP), ensure its suitability.

On-site generation can reduce oxygen supply costs by an estimated **50–90%** [4], as it circumvents the costs associated with production, liquefaction, transportation, and cylinder rental of bulk oxygen. These savings directly offset a portion of the electrolyzer's operational expenses, effectively lowering the net cost of hydrogen when the oxygen byproduct is valorized [5].

For investors, the electrolyzer becomes a dual-revenue (or dual cost-avoidance) asset, improving the overall return on investment while providing an invaluable, life-saving, uninterrupted oxygen supply.

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3. HYDROGEN FOR MRI SUPER-COOLING: A FUTURE-PROOFING STRATEGY

Magnetic Resonance Imaging (MRI) machines utilize superconducting magnets requiring cryogenic temperatures (~4 K, −269°C), traditionally achieved using liquid helium — a scarce, non-renewable, and increasingly expensive resource [6]. Global helium shortages have already impacted hospital operations, causing downtime and escalating costs [6].

Hydrogen is emerging as a viable alternative cryogenic coolant. Research demonstrates its efficacy, particularly with newer high-temperature superconductors like MgB_2 [7]. Engineering studies are exploring MRI magnet designs incorporating hydrogen cooling loops, signaling a shift away from helium [7].

While direct hydrogen-cooled MRI systems are still developmental, they could utilize liquid hydrogen or closed-cycle hydrogen gas refrigeration. This transition promises to drastically reduce or eliminate helium consumption, yielding

substantial operational cost savings and mitigating supply chain vulnerabilities.

From an investment perspective, early adoption of hydrogen-based cooling technologies, synergizing with on-site hydrogen production, positions a hospital as a leader in operational efficiency and reliability, addressing the critical and worsening issue of helium scarcity [6].

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4. HYDROGEN IN HOSPITAL RESEARCH AND ADVANCED MEDICAL APPLICATIONS

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Large hospitals are often centers for research and specialized treatments, creating diverse opportunities for hydrogen utilization.

- **Laboratory operations**: Many analytical instruments, such as gas chromatographs, traditionally use helium as a carrier gas. Due to helium scarcity and cost, laboratories are increasingly transitioning to hydrogen, which can be supplied reliably and cost-effectively from an on-site source, eliminating bottled gas expenses and improving safety by enabling low-pressure piped supply [6].
- **Therapeutic potential**: Hydrogen gas itself is being investigated as a therapeutic agent with antioxidant and anti-inflammatory properties. Over **2,000 scientific publications** and numerous human trials are exploring its application in conditions ranging from cardiovascular and neurodegenerative diseases to tissue injury treatment [7]. A hospital with integrated hydrogen infrastructure would be uniquely positioned to participate in, or lead, such cutting-edge clinical trials, attracting research funding and enhancing its reputation as an innovation hub.

For investors, these applications represent significant upside. They leverage existing hydrogen infrastructure at a low marginal cost for potentially high returns in research capabilities, grant attraction, and pioneering new treatments.

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5. HYDROGEN FOR ZERO-EMISSION BACKUP POWER AND ENHANCED RESILIENCE

Uninterruptible power is non-negotiable in hospitals. Traditional diesel generators, while common, suffer from emissions, maintenance burdens, and potential unreliability, as evidenced during major outages like Superstorm Sandy where many failed [8].

Green hydrogen, through fuel cells or hydrogen-fueled generators, offers a clean, highly reliable alternative. Several U.S. hospitals already employ fuel cells for critical backup power, demonstrating superior resilience and eliminating local air pollution [8].

A hydrogen-powered backup system typically comprises on-site hydrogen storage and fuel cell units sized for essential loads. These systems can provide continuous power (cogeneration) during normal operations and instantly ramp up during outages, offering extended runtime limited only by stored hydrogen volume.

Financially, while CapEx for fuel cells may be higher than diesel generators, long-term operational savings accrue from eliminated diesel fuel costs, reduced maintenance, and potential carbon pricing avoidance. Furthermore, fuel cell installations have been eligible for incentives like Investment Tax Credits and Renewable Energy Credits [8], improving project economics.

The U.S. Department of Energy's **$9.5 billion program** for clean hydrogen hubs signals strong policy support [9]. By ensuring operational continuity and mitigating risks associated with power failures, hydrogen backup systems protect a hospital's core mission and financial stability [10].

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6. PEAK SHAVING AND ENERGY COST OPTIMIZATION WITH HYDROGEN

Hospitals, as 24/7 high-energy consumers, incur substantial electricity costs, including significant peak demand charges. An integrated hydrogen system enables peak shaving: producing and storing hydrogen during off-peak hours (when electricity is cheaper or surplus on-site renewables are available) and using it to generate power via fuel cells during high-cost peak periods.

This flattens the hospital's demand profile, reducing energy procurement at premium rates and lowering demand charges [11]. This strategy, proven in other sectors [11], can yield significant annual savings.

For example, shifting even a fraction of peak load consumption (e.g., from **$0.20/kWh peak** to **$0.10/kWh off-peak** equivalent via hydrogen) can accumulate to hundreds of thousands of dollars saved annually for a large facility. The hospital's energy management system can automate this process, responding to real-time pricing or participating in demand-response programs for additional revenue.

The round-trip efficiency of power-to-hydrogen-to-power is lower than batteries, but hydrogen offers superior scalability for long-duration storage and multi-use versatility, making it an active energy optimization asset that can reduce a hospital's share of the estimated **$6 billion annual energy expenditure** across U.S. hospitals [12].

7. ECONOMIC ANALYSIS AND FINANCIAL VIABILITY

Integrating green hydrogen involves significant CapEx for electrolyzers, storage, fuel cells, and auxiliary systems, potentially ranging from **$5–10 million for a 1 MW scale system**. However, this can be mitigated through phased deployment or innovative financing like Power Purchase Agreements (PPAs) or Energy-as-a-Service models, shifting CapEx off the hospital's balance sheet [12, 8].

The OpEx, primarily electricity for electrolysis, is countered by substantial savings and revenue offsets:

- **Oxygen cost savings**: Up to **50–90% reduction** in oxygen supply costs [4], potentially saving hundreds of thousands annually and effectively subsidizing hydrogen production [5].
- **Fuel displacement**: Elimination of diesel fuel costs for backup generators and reduced grid electricity purchases.

- **Energy cost optimization**: Tangible savings from peak shaving and potential demand response revenue.
- **Incentives and carbon credits**: Access to grants, tax credits (e.g., U.S. **30% ITC** for fuel cells [8]), renewable energy credits, and potential future monetization of carbon reductions.
- **Reduced maintenance and enhanced reliability**: Lower maintenance for fuel cells compared to diesel engines and avoidance of outage-related financial and reputational losses.

While current green hydrogen costs (2–3 times fossil-derived [13]) may suggest longer payback periods (e.g., **7–10 years**), costs are projected to fall significantly. IRENA forecasts green hydrogen **below $2/kg within the next decade**, and electrolyzer CapEx to drop by **40% short-term** and **80% long-term** [13]. This trajectory ensures improving project economics over the system's lifespan.

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The "value stacking" approach — utilizing the hydrogen system for multiple concurrent applications — maximizes asset utilization and return on investment.

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8. NAVIGATING THE REGULATORY LANDSCAPE

Successful implementation requires adherence to several regulatory frameworks:

- **Medical oxygen production**: Compliance with health authority regulations (e.g., FDA, EMA) for purity, GMP, labeling, and potentially manufacturing licenses [3]. These are established protocols already followed by hospitals with on-site PSA oxygen generators.
- **Hydrogen storage and safety**: Adherence to codes like NFPA 2 and NFPA 55 in the U.S. [14], involving engineered safety systems (ventilation, leak detection, emergency shut-offs), hazard analyses, and staff training. These are standard in industrial settings.
- **Electrical and building integration**: Compliance with emergency power standards (e.g., NFPA 110), building codes, and environmental air quality regulations (where hydrogen excels by eliminating local pollutants).
- **Future trends**: Proactive engagement with evolving hydrogen regulations can position the hospital as a leader and benefit from supportive policies.

While comprehensive, these regulatory requirements are manageable and ensure safe, compliant operation, creating a certified, trusted system [1, 3].

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9. CONCLUSION: INVESTMENT OUTLOOK AND STRATEGIC LONG-TERM VALUE

The integration of green hydrogen in large hospitals transcends a mere sustainability initiative; it is a strategic investment in future-proof infrastructure that delivers energy security, operational cost control, enhanced resilience, and potential new service avenues. For investors, this presents a compelling opportunity characterized by:

- **Diversified value streams**: Revenue and savings from oxygen, power, heating, backup, and research.
- **Significant ROI potential**: Driven by cost savings, operational efficiencies, and declining technology costs, particularly with value stacking [5].
- **Enhanced risk mitigation**: Insulation from fuel supply shocks, power outages, and price volatility.
- **Strong ESG credentials**: Demonstrable emissions reductions and improved local air quality, enhancing brand equity and attractiveness to ESG-focused capital [10, 8].
- **Scalability and market impact**: A replicable model for a multi-billion dollar global market in healthcare decarbonization.
- **Policy alignment**: Synergy with government decarbonization and hydrogen strategies, positioning for future incentives and regulatory advantages [10].

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Investing in hospital-based green hydrogen systems offers a unique confluence of financial return, operational improvement, and profound societal impact. It transforms hospitals into resilient, efficient, and environmentally responsible institutions, aligning profit with purpose in a critical sector.

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